

Sensor Calibration and Positioning System for Maritime Application



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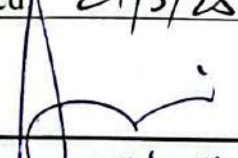
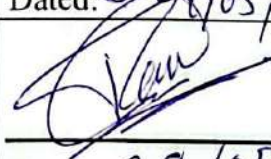
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We dedicate this project to our respected supervisor for their invaluable guidance, and to our parents and siblings, whose unwavering support and encouragement have inspired us to strive beyond limits.

Abstract

This project focuses on the design and development of an enhanced sensor system for maritime applications, aimed at overcoming the limitations of a previously manual-controlled setup. The initial provided prototype, which provided three degrees of freedom—linear translation, depth adjustment, and rotational control—lacked any form of analog or digital feedback. As a result, it suffered from poor accuracy and repeatability, with depth measurements deviating by up to 6 in, unreliable angular control, and no defined precision for linear motion. In response, a new, compact, and efficient system has been developed, integrating modern sensors and electronics to enable real-time data acquisition and providing control for all three degrees of freedom. The upgraded system includes safety mechanisms, improved mechanical design, and a user-friendly interface, achieving a significant increase in accuracy, up to 1 mm, making it more suitable for critical maritime operations. This development not only enhances operational reliability but also provides a solid foundation for further automation and remote-control capabilities.

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Nomenclature

HMI	Human Machine Interface
Al	Aluminum
DAQ	Data Acquisition
PWM	Pulse width modulator
MS	Mild Steel
SONAR	Sound Navigation and Ranging
GUI	Graphical User Interface
PU	Polyurethane
ID	Internal Diameter
OD	Outer Diameter

CHAPTER 1: INTRODUCTION

The research work done in this thesis is presented in four parts. The first part covers the motivation, problem statement and objective of the project. The second part covers the background, literature review and the aim of this entire thesis. The third and fourth part covers the methodology, in which the third part explains the mechanical portion, the components used, the experimental setup utilized and working of the mechanical system while the fourth part covers the Instrumentation portion of the system i.e., software of the system, working, and interface. Finally, the last part presents the conclusion of this research.

1.1 Motivation

The motivation for this project is to address the limitations of existing SONAR positioning systems, which rely on manual operation and lack precision, efficiency, and safety features. Due to limited local research in this field, current systems are slow, error-prone, and unsuitable for the high accuracy required in mission-critical applications. This project aims to develop an automated, precise, and locally manufactured SONAR platform with integrated sensors, digital control, and safety mechanisms. By enhancing accuracy, efficiency, and reliability, the system will modernize the calibration process and will contribute to strengthening local capabilities in critical sensing technologies.

1.2 Problem Statement

As the primary sensory system for navigation and detection, SONAR requires extraordinary precision and accuracy to function effectively. However, due to the country's limited research and competence in this field, there are several major challenges that need to be addressed. The current system relies on manual positioning and control methods, which limit its efficiency. Additionally, the process lacks the necessary speed and accuracy for optimal sensor calibration because of the absence of feedback mechanisms such as encoders with data acquisition. As a result, SONAR systems require extensive and time-consuming calibration before deployment, resulting in inefficacy of the system. For effective operation, each spatial calibration must be both quick and precise to ensure effectiveness. Furthermore, despite being mission- and business-critical equipment, the existing

system lacks any safety features to handle anomalies, posing a significant risk during operation.

1.3 Objective

The objective of this project is to develop a precise and accurate SONAR positioning platform using local manufacturing capabilities. The current prototype will address existing challenges by ensuring stable positioning in dynamic underwater environments. It will feature remote/ computer controlled On-rail movement and Vertical translation, with 1 mm precision and 1-degree rotational accuracy. Additionally, a user-friendly human-machine interface (HMI) will be developed for easy control of all the on-board equipment and ultimately the platform.

CHAPTER 2: BACKGROUND

2.1 Background and Literature Review

The advent of position sensors such as RADAR, SONAR, LIDAR among others have had a considerable impact on the navigation prospect. Such sensors have been the center of major innovations in different fields both in civilian and in defense applications, such as maritime applications, air, satellite, land and R&D, ever since their advent over more than 100 years ago. Sensors such as Radars and Sonars are mounted onboard of smart and flexible platforms such as large ships, submarines, unmanned vehicles, aircrafts, and in many maritime R&D setups. These technologies focus on remote detection of target, and both may encounter many common scientific challenges. One of the major problems being its calibration or positioning due to its mounting system or platform. [1]

Being sensitive and critical positioning sensors, the calibration and proper positioning of the sensors is of utmost importance for the best noise free results, and as a result they require a system that would provide them with such a platform, so that the sensors are properly calibrated and positioned, before they are used. Calibration aligns the sensor's output with known standards, ensuring accuracy and reliability. If a sensor is not mounted securely or is exposed to external vibrations, misalignments, or environmental changes, its calibration can drift, leading to inaccurate readings, and thus accurate calibration depends on stable setup. A well-designed platform ensures that sensors remain in a fixed, known position relative to

the system and to each other. This consistency is vital for repeatable, reliable measurements—especially in navigation, where even minor shifts can cause significant errors. To accommodate these calibration and positioning challenges, a system was designed which would control the position of the concerned sensor in 3 degrees of freedom, i.e. the vertical movement which controls the depth, the rotary movement which controls the angular position, and the linear position which controls the movement along the platform.

The current system, while functional, faces several limitations that hinder its performance and precision. Primarily, it operates manually, lacks integrated sensors for feedback or monitoring, and offers low positional accuracy. One of the key components being the mechanism for the vertical movement of the sensor, relies on an aluminum extrusion measuring 40 mm × 40 mm × 670 mm. This extrusion includes holes spaced at 150 mm (approximately 6 in) intervals along its length, which allow for discrete height adjustments. The vertical positioning is done manually, using an overhead crane to move the extrusion up or down. Once the desired height is reached, a pin or rod is inserted into the nearest hole to lock the position in place. While this mechanism provided a simple and sturdy method for vertical adjustment, it inherently limited the system's accuracy in the vertical direction to increments of 150 mm. As a result, fine positioning of the height for greater precision applications was not possible with this setup.

For translation along 10m long rails (on-rail movement), the system was equipped with small wheels that allowed it to roll along a set of rails. When repositioning was required, the crane would lift the system slightly to reduce weight on the wheels, after which an operator would manually push or pull the system along the tracks to the desired location. While the wheels provided some convenience by making the system easier to move when lifted, they did not offer any form of precision control. This approach was limited because of no sensors to provide feedback data to accurately determine the systems position along the rail. Without real-time positional data, it was not possible to precisely assess or repeat the system's location, making it unsuitable for such system that demand fine control and high positional accuracy.

Finally, for the rotary motion, a cylinder with Teflon embedded in the tower-flange section is integrated. The rod would pass through this Teflon embedded cylinder. Below the cylinder, a timing pulley was attached, which was rotated via belt through

NEMA-34 stepper motor. The limitation of the rotation was attributed to the use of Teflon instead of bearing assembly, which would be a nuisance to rotation, and like other 2 degree of freedoms, there was no digital output to get good accuracy to stop at the desired angle. Moreover, the reading in the cylinder was to be taken from an angular indicator.

The sensor attached is a mission critical instrument and its safety is of utmost importance. The current system does not have any safety measures installed in it that would accommodate its safety in worst case scenarios.

2.2 Aim of the Project

The aim of the project is to automate it, make it possible to remotely operate all the onboard equipment to set the position of the sensor, increasing the accuracy and precision of the system by integrating sensors, feedback systems and high accuracy motors. Installation of safety features such as limit switches to turn off the machine in case of mishap, pulse detector safety circuit in case of connection breakout. The finalized product should be capable of:

1. Remote controlled system
2. Platform rotation with 1-degree precision.
3. Vertical translation and accurate on-rail movement with 1-mm precision.
4. A user-friendly HMI for control and data acquisition.
5. Safety features integrated into the system



Figure 1: Old Model

CHAPTER 3: MECHANICAL DESIGN SETUP

This chapter presents the mechanical setup designed to enable precise positioning and movement of the SONAR sensor system. The structure is divided into three key subsystems: depth control, rotational control, and linear (on-rail) movement, each engineered for smooth, stable, and accurate operation. All three sub systems are made up of MS. The depth control system allows controlled vertical adjustment of the sensor, essential for reliable depth readings, while the rotational control mechanism enables precise angular positioning for effective area scanning. Complementing these is the linear movement system, which facilitates smooth traversal of platform along the rails to cover broader scanning zones. Together, these integrated mechanical components form a cohesive system optimized for high-performance SONAR operations.

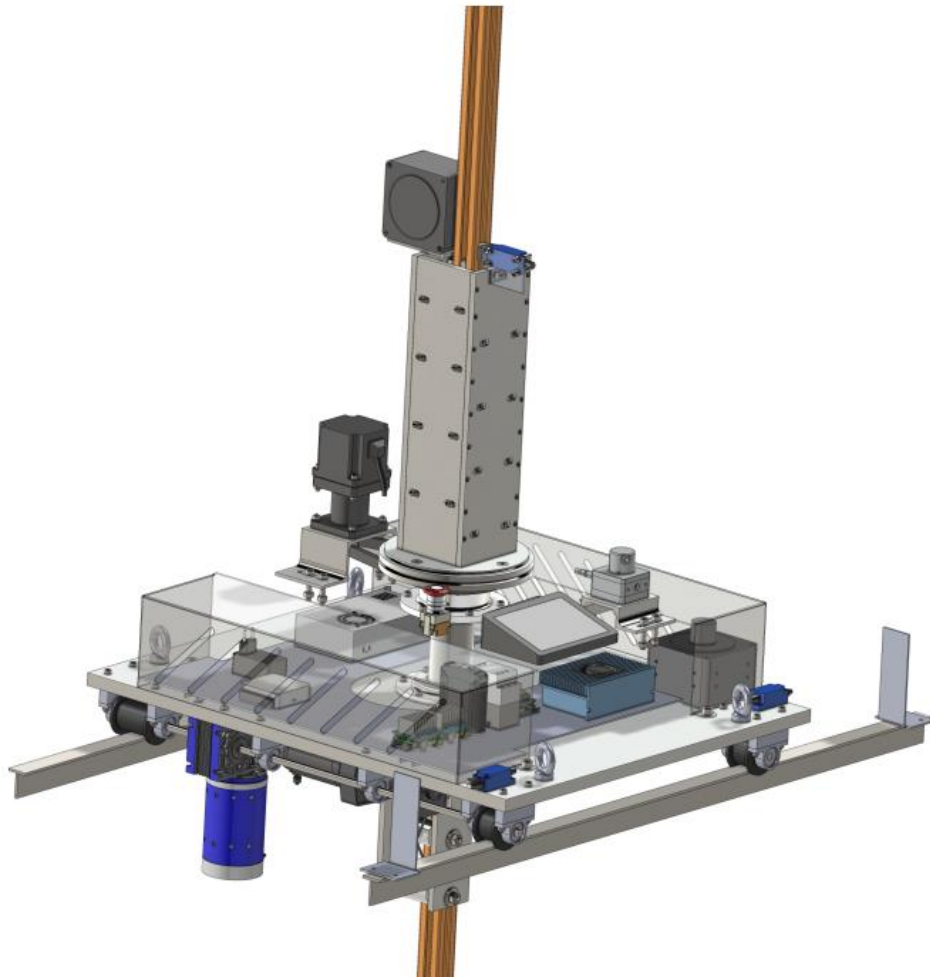


Figure 2: Final Assembled Model 

3.1 Depth Control Setup

To enable precise and smooth vertical movement of the SONAR sensor during operation, a dedicated depth control system has been implemented. This system is essential for lowering the sensor into the water with controlled motion, ensuring accurate depth positioning required for reliable SONAR readings. To achieve smooth and stable movement, the system employs a guided mechanism where rollers reduce friction and prevent lateral shifts. A motorized winch controls the vertical travel of the sensor, providing consistent and adjustable motion while maintaining tension in the system. Together, the rollers and winch ensure that the sensor descends and ascends without any sudden jerks or instability, which is critical for both calibration and operational accuracy.

The depth control system consists of following parts:

1. Al Extruded Rod
2. Vertical Tower with Rollers
3. Driving Winch



Figure 3: Depth Control Assembly

3.1.1 Al Extruded Rod:

To control the depth of the sensor, an Al extrusion of 40 mm x 40 mm profile is used. The total length requirement was 22 feet or 6.7 m. The Al extrusions used in this case are 6 ft long (approx. 1.8m). To achieve the desired length custom linkages keys were designed, that were used to join two pieces. In total, four extrusions were connected to form a continuous assembly measuring 24 feet (7.3 meters), slightly exceeding the initial requirement for added flexibility. The sensor is mounted at the end of this extrusion assembly, and a hook has been integrated to securely attach the depth control winch.

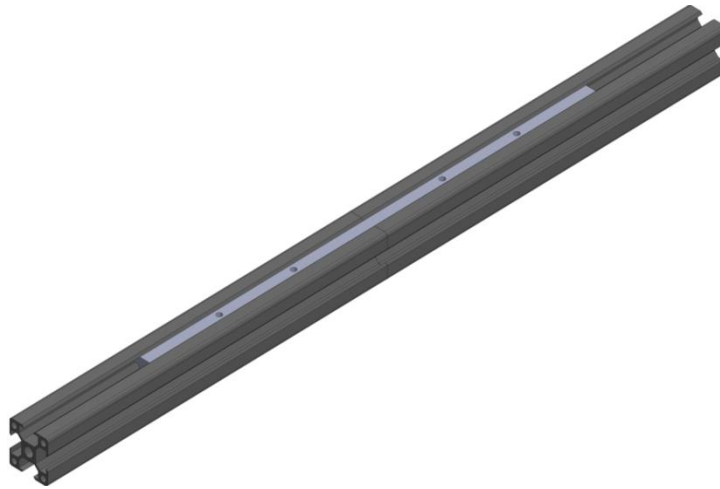


Figure 4: Assembled Extrusions

3.1.2 Vertical Tower with Rollers:

The rod passes through a vertical tower fitted with PU rollers to ensure guidance and precision for vertical motion. A total of 16 rollers are deployed within the tower rack, arranged in opposing sets of 8 on each parallel side to provide balanced support. The rod fulfills three primary functions:

1. It maintains precise axial alignment at the center of the tower.
2. It constrains lateral displacement, effectively preventing movement along the X and Y axes.
3. It facilitates smooth and stable vertical translation, ensuring controlled depth adjustment

3.1.3 Driving Winch

The AI extrusion is moved by an industrial-grade, motorized winch [2] rated for 2000 lbs of load capacity. The winch is connected to the extrusion by a hook. A 10.5-meter-long steel cable is used to transmit motion from the winch to the extrusion. The extrusion assembly is mounted on a block, which is mechanically linked to the underside of a rotating cylinder, providing structural support and allowing controlled vertical movement during operation.

Components	Specification	Function
Aluminium Rod	4x6ft with 40x40mm profile	Provides attachment to SONAR for required vertical displacement
Tower Rack Guide rollers	16x PU-based rollers with 30mm dia and 50mm length	Provide vertical guidance for rod during vertical translation
Winch	Capacity: 2000 lbs, 12V	For vertical translation of the rod, regulated via encoders

Table 1: Summary of depth control setup

3.2 Rotation Control Setup

To enable controlled rotation of the system, a dedicated rotational drive mechanism has been designed. This setup is essential for precise angular positioning of the sensor assembly, ensuring accurate scanning during SONAR operations. The system employs a long, hollow cylindrical flange that passes through both the upper and lower support plates. Bearings are installed in each plate interface to minimize friction and enable smooth, stable rotation. A timing pulley mounted at the top end of the cylinder is driven by a NEMA-34 stepper motor, providing consistent and adjustable rotational motion while maintaining alignment and load stability. Together, the bearings and motor-driven pulley ensure that the cylinder rotates without any jerks or misalignment, which is critical for both calibration and operational performance.

The rotational drive system consists of the following parts:

1. Rotating Cylinder

2. Bearings
3. Stepper Motor (NEMA-34)



Figure 5: Rotation Setup

3.2.1 Rotating Cylinder

The Al extrusion first passes through the vertical tower equipped with rollers and then continues through the rotating cylinder. The cylinder features an internal diameter of 59 mm, providing a compact and efficient design while accommodating smooth passage of the extrusion. At the lower section of the cylinder, two mounting holes are provided for securing the winch block, ensuring stable attachment during operation. At the top end of cylinder in connection with the tower rack, a 120-teeth 3M pulley is installed, which enables precise rotational control of the system when driven by the stepper motor. The vertical tower is mounted directly on top of this rotating cylinder, integrating the depth control and rotation mechanisms into a cohesive assembly.

3.2.2 Bearings

Two types of bearings are utilized in the assembly to ensure smooth and stable operation: Two SKF 61813 radial bearings [3] and one SKF 32913 thrust bearing [4]. The thrust bearing is specifically employed to support and lift the vertical load

concentrated on the rotating cylinder and provide stability in case of any stray external axial loading on the system during operation, while the radial bearings facilitate smooth, low-friction rotation. Custom bearing housing has been fabricated to securely enclose the bearings, protecting them from mechanical damage and environmental exposure. The thrust bearing, along with one radial bearing, is installed at the upper plate, providing both axial load support and rotational guidance. The lower plate houses the second radial bearing, which further stabilizes the rotation and maintains alignment.

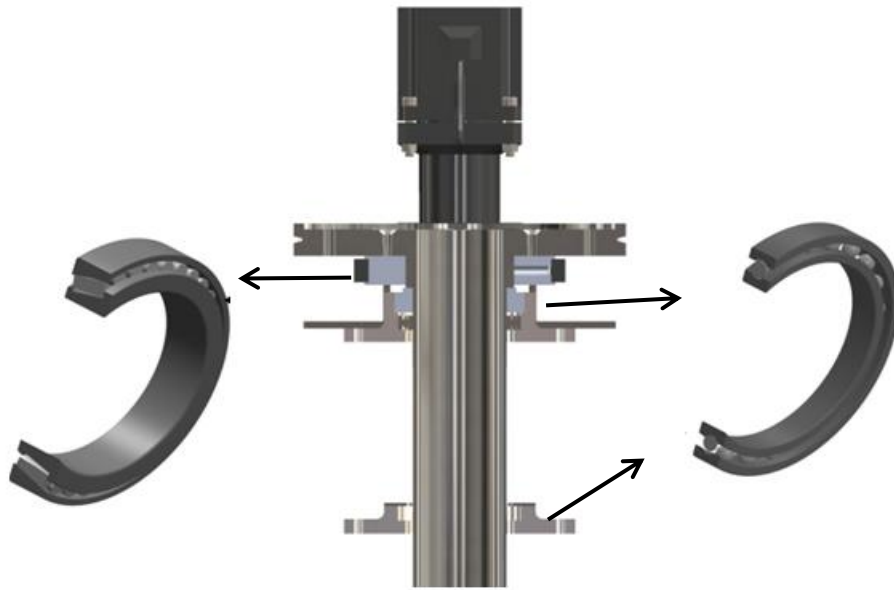


Figure 6: Bearings, Thrust (left installed at top), Radial (right)

3.2.3 Stepper Motor (NEMA 34)

A NEMA 34 stepper motor, rated at 4.5 N·m base torque, is employed to drive the rotational system. The motor is coupled with a 6:1 reduction gearbox [5] to enhance torque output and ensure precise speed control. A 50-teeth 3M pulley is mounted on the motor's output shaft, which drives a 120-teeth 3M pulley installed at the top of the rotating cylinder. The operation of the stepper motor is controlled via DMA-860H stepper motor driver through LABView instructions provided by DAQ Card. The rotational motion is regulated via 2m linear encoder that has its linear reading translated into degrees

The rotational speed is determined by the pulley ratio:

$$\text{Speed Ratio} = \frac{\text{Driving Teeth}}{\text{Driven Teeth}} = \frac{5}{120} = 0.4167$$

Given the input speed of 14.4 rpm from the motor shaft, the output speed at the rotating cylinder is:

$$\text{Output Speed} = 14.4 \text{ rpm} \times 0.4167 = 6 \text{ rpm}$$

Thus, the system achieves a controlled rotational speed of 6 rpm at the sensor assembly, ensuring smooth and accurate angular positioning during operation.

Components	Specification	Function
Rotating Cylinder	Mild Steel based flange with 59mm ID and varying OD (stepped OD) at locations	Main rotation flange assembly connecting bearings and platform base plates
Bearings	SKF 61813 (65mm ID, 85mm OD, width 10mm)	Responsible for bearing radial loading and allowing rotational motion of the vertical motion assembly
	SKF 32913 (65mm ID, 90mm OD, width 17mm)	
NEMA-34 Stepper motor	Base Torque: 4.5 Nm, 6:1 Gearbox, DMA-860H Driver	Main element of rotational assembly, controlled through LABView panel and regulated by encoder

Table 2: Summary of depth control setup

3.3 Linear Movement

The on-rail movement system consists of four wheels and a DC motor-driven mechanism. Linear motion along the rail is achieved using an 1800 rpm DC motor coupled with a 60:1 reduction gearbox [6], resulting in an output speed of 30 rpm for controlled and stable travel. The motor is connected to a long drive shaft that runs along the front side of the system. This shaft is supported by a series of pillow block bearings—two of which ensure smooth, low-friction rotation. Power transmission is accomplished through a keyed connection between the motor and the shaft. As the shaft rotates, it directly drives the front pair of wheels, which in turn

engage and propel the rear wheels, allowing the entire system to move smoothly and synchronously along the rail.

Components	Specification	Function
Wheels	Mild Steel wheels with 11.6mm ID with varying lengths based on requirement (rear-end with smaller wheels and larger shaft, front-end larger wheel for on-track adjustment)	Responsible for on-rail movement.
Drive Shaft	Rear-end with two smaller shafts and Front-end with longer 11.6mm ID drive shaft with keyed connection at motor attachment	Responsible for connecting the wheels to the platform and motor for translation
DC Stepper Motor	30 rpm reduced torque, 60:1 gearbox, 24V	Main element responsible for moving wheel assembly

Table 3: Summary of On-rail linear movement control setup

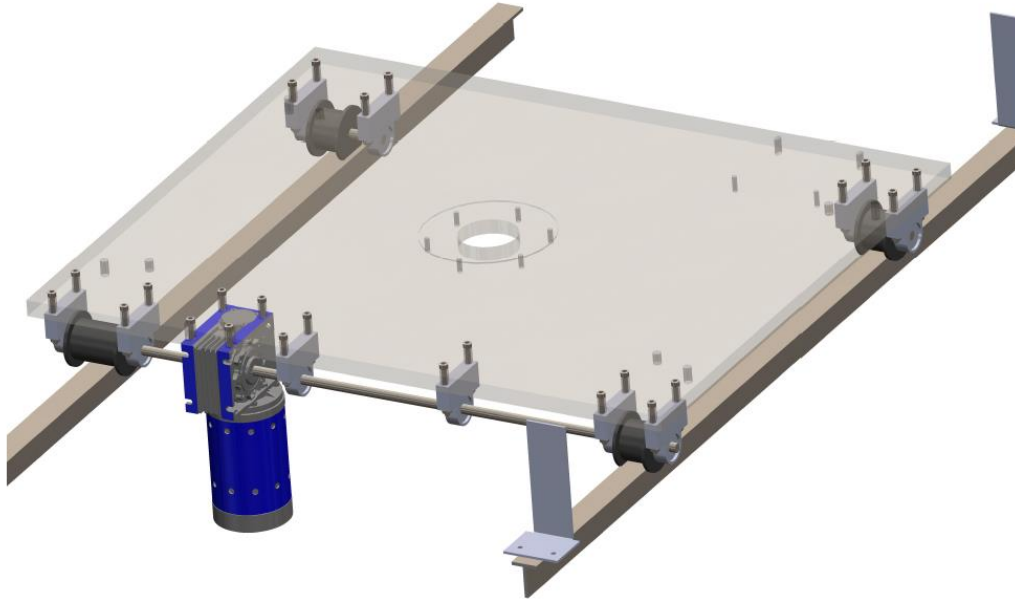


Figure 7: On-rail Motion Setup

CHAPTER 4: ELECTRONICS AND SOFTWARE

This chapter outlines the technical methodology employed in designing and implementing the control system for the platform, focusing on both the hardware and software aspects. The goal was to create a robust and reliable system capable of performing complex motion control tasks with high accuracy and integrated safety. The system merges human-machine interaction principles with embedded electronics and software logic, offering an efficient interface for both local and remote operations.

The control architecture is modular, scalable, and designed with redundancy and fault-tolerance in mind. Emphasis has been placed on ensuring seamless communication, effective actuation, and emergency response via software logic and physical components such as limit switches and a pulse check circuit.

4.1 Circuit Overview

The foundational design principle was to maintain uninterrupted, real-time communication between the on-job operator (Master control), control room operator (Slave control) and the mobile trolley platform. The control system comprises a

DAQ (Data Acquisition) interface, motor drivers, encoder feedback, and safety systems that work synchronously through an Ethernet-based communication line.

Figure 7 illustrates the complete electrical and control wiring schematic. It highlights key components including power lines, signal pathways, safety interlocks, and data lines:

1. **Red lines** represent high-voltage AC power delivery (Live wires).
2. **Green lines** denote pulse or signal pathways.
3. **Blue lines** are Ethernet-based data communication routes.
4. **Purple lines** represent emergency safety triggers through limit switches.

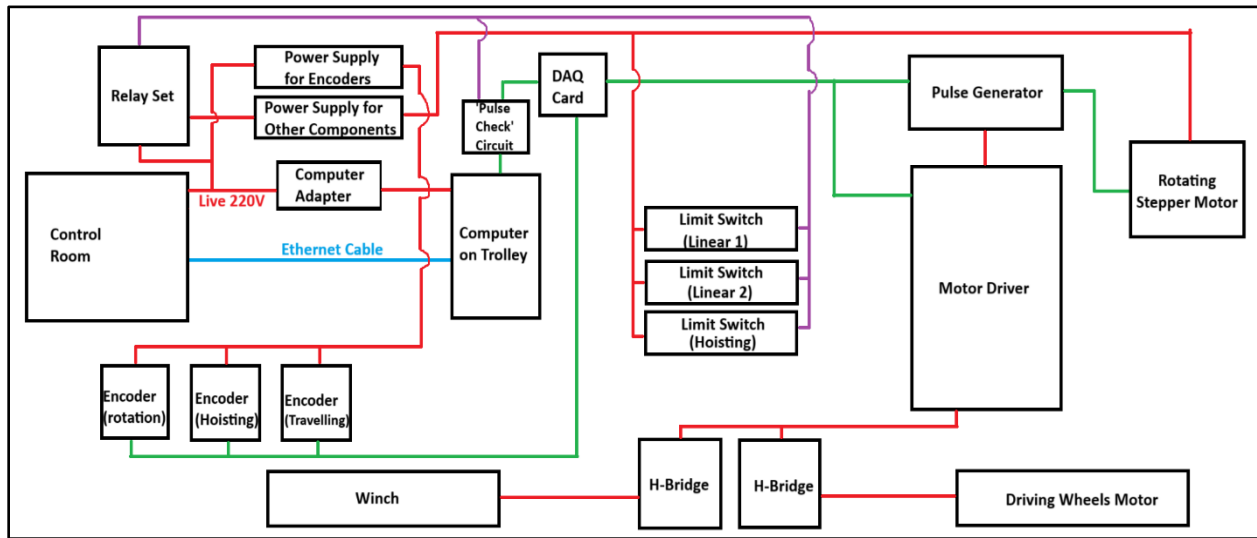


Figure 8: The General Overview of the circuit.

4.2 Wiring Architecture

The wiring network originates in the control room, where a 220V AC mains supply and an Ethernet cable are routed toward the trolley unit. Ethernet was chosen after extensive latency testing and was found to have an average signal delay of approximately 100 milliseconds, which falls well within acceptable thresholds for control applications. Furthermore, Ethernet demonstrated negligible signal attenuation over longer cable lengths, making it ideal for large-scale systems.

The system is initiated via a master switch located in the control room. However, the design incorporates multiple automatic fail-safes. The system powers down entirely if:

1. Communication to the DAQ is interrupted.
2. Any of the integrated limit switches are triggered.

The platform's rotary encoders, critical for position feedback, are connected to a dedicated power source. This ensures they continue to function even when the main system is shut down due to an emergency, allowing the system to resume operation with accurate position memory upon restart.

4.3 Motor Actuation and Driving Logic

Three distinct motors are used in the platform:

1. A **stepper motor** for rotational motion.
2. A **DC winch motor** for vertical hoisting.
3. A **DC motor** for trolley locomotion.

Each motor requires a different control strategy:

1. **Pulse Width Modulation (PWM)** is used to drive the stepper motor. PWM control provides fine resolution and precise rotational control, which is vital for sonar orientation.
2. **H-Bridge circuits** are used to control the DC motors. Two independent H-Bridge drivers enable bidirectional motion and speed regulation of the trolley and winch motors.



Figure 9: H-Bridge (left), Pulse Generation Module (right)

4.4 Data Acquisition

For acquiring precise positional data from the mechanical system, wire pull encoders are used. These encoders provide real-time feedback on the linear, rotational, and depth movements of the assembly. To ensure uninterrupted operation, even during power fluctuations, the encoders are powered by a separate 24V supply independent of the main system. The positional signals from the encoders are transmitted to a Data Acquisition (DAQ) card, which processes the raw signals and converts them into readable position outputs in meters or degrees on the display interface. This setup enables accurate monitoring and control of the system's movements during operation.

4.4.1 Encoders:

A total of three pull wire encoders [7] are integrated into the system, each selected based on the range and type of movement they are meant to measure:

1. **Linear Movement Encoder:** A 10-meter range encoder is used to track the on-rail linear motion of the entire setup. This allows for precise measurement across the full length of the scanning rail. It is mounted on the base plate.
2. **Depth Control Encoder:** Another 10-meter range encoder is dedicated to measuring the vertical movement of the SONAR sensor, ensuring accurate depth positioning during deployment. It is mounted on top of the vertical tower,
3. **Rotational Encoder:** A 2-meter range encoder is used to monitor the rotational movement of the sensor assembly. Though physically smaller in

range, it provides high-resolution angular position data required for accurate SONAR scanning. It is mounted on a bracket designed for the encoder on the upper plate.



Figure 10: Pull Wire Encoder

4.4.2 DAQ Card

The positional data from the encoders is captured and processed using a National Instruments (NI) USB DAQ card, as shown in the figure. This particular DAQ card is designed for versatile signal acquisition with USB connectivity, making it ideal for portable and PC-based measurement systems. It features multiple analog and digital input channels, allowing simultaneous connection of the three encoders. The card receives voltage or pulse signals from the encoders and, using internal analog-to-digital converters (ADC), translates them into digital position readings. The processed data is then displayed in engineering units such as meters or degrees via software interfaces like NI LabVIEW. This real-time feedback loop enables operators to monitor and control the system's position accurately during operation.



Figure 11: NI DAQ Card

4.5 Software Control Interface

The system utilizes **LabVIEW** for software-based control and monitoring. A GUI allows operators to visualize real-time data and manage commands with ease. The system enforces spatial boundaries for each actuator to ensure safe operation:

1. **Hoisting Range:** 0 to 6 meters.
2. **Trolley Travel Range:** 0 to 12 feet.
3. **Sonar Rotation:** -180° to $+180^{\circ}$.

These bounds are hardcoded into the LabVIEW software to prevent mechanical overextension. The control software runs continuously during operation and serves as the primary interface unless overridden by limit switches.

Figure 9 presents a LabVIEW interface that displays real-time positional data and control parameters of the system.



Figure 12: Software User Interface

4.5.1 Master-Slave Mode Switching

A key feature of the system is its dual-master capability, which allows operation either from the trolley or the remote-control room:

1. When the operator is physically on the trolley, the onboard computer becomes the master controller.
2. When control shifts back to the control room, the system recognizes the control room computer as the master.
3. When both operators try to operate the system, the system recognizes the on-board operator as master based on location preference set in the code

This flexible architecture is achieved via dynamic TCP/IP switching. Initially, the LabVIEW interface uses localhost as the default target IP. For real-world deployment, this must be changed to a static IP address. Static IP assignment is performed in the system's IPv4 settings and must be complemented by:

1. Matching subnet masks.
2. Enabling network discovery.
3. Configuring firewall exceptions to allow LabVIEW through.

This configuration enables the platform to be scaled, allowing one control room to command multiple trolleys operating simultaneously in distributed environments.

Importantly, if communication is lost during command transmission, the system is designed not to execute the last received command, thereby preventing unintended motion or potential accidents.

4.6 Limit Switches

The system uses limit switches as a safety mechanism when the system reaches end boundaries in different movement directions. Two types of encoders are used in the system:

1. **Push-type switches [ME-8111]** are placed at either end of the trolley track. When activated, they interrupt the relay line, cutting power to all motors.
2. **Rotary-type switch [ME-8104]** is installed at the top of vertical tower. The extrusion has a switch engager, that engages the switch. This provides end-stop detection for vertical hoisting.

All limit switches directly engage relay cutoffs rather than software logic, ensuring immediate system stoppage during critical conditions.



Figure 13: Limit Switch: Rotary Type (Left) & Push Type (Right)

4.7 Pulse Check Circuit

To enhance safety, a **pulse check circuit** based on the 555 timer is employed. This circuit functions as a watchdog, monitoring pulse continuity from the DAQ system.

- The 555 timer is configured in a **monostable or re-triggerable mode**, forming a Schmitt trigger-type arrangement.
- If no pulse is detected within a specified timeout window (typically 2–3 seconds), the timer triggers a shutdown sequence by disabling the main power relay.

This circuit provides a hardware-based safety net that operates independently of software, adding redundancy to the system’s fault-tolerance capabilities.

CHAPTER 5: RESULTS AND DISCUSSION

After completion of the mechanical design and subsystem integration, the SONAR positioning platform was fully assembled and dispatched for field testing. This marked the transition from controlled laboratory trials to real-world operational scenarios, providing the first opportunity to evaluate system performance under authentic environmental conditions. Prior to deployment, extensive structural simulations were conducted, confirming that the machine could comfortably withstand anticipated loads without exhibiting concerning deflections, bending, or structural fatigue. This validation established confidence in the platform's long-term mechanical resilience, particularly during repeated depth hoisting and rotational operations in waterborne environments.

During assembly, special emphasis was placed on safety and operational integrity. Limit switches and mechanical stoppers were systematically installed on all movement axes — linear, rotational, and vertical — serving as critical safeguards. These physical limiters were designed to halt motion immediately upon reaching predefined boundaries, preventing accidental overextension and protecting both the equipment and the operator. Each component integrated into the machine, including motors, gearboxes, bearings, and sensors, underwent rigorous lab testing and calibration. All components consistently achieved their intended performance targets, indicating full readiness for cohesive system trials.

A key highlight of this development phase was the successful integration of a customized LABView-based Human-Machine Interface (HMI). This centralized control system allowed operators to seamlessly manage the three core movement subsystems — depth control, rotation, and linear traversal — through a user-friendly interface. Ethernet-based communication enabled remote operation from the control

room with minimal latency, while onboard manual control remained available via dynamic master-slave mode switching. This flexible control architecture provided significant operational convenience, particularly during field deployment and maintenance.

Precision testing of the platform yielded highly encouraging results. Initial lab tests using Arduino UNO for individual component verification and LABView for integrated system trials consistently achieved a translational accuracy of approximately 1 mm along both linear and vertical axes. Rotational subsystem trials maintained angular accuracy within ± 1 degree, affirming suitability for precision SONAR scanning. The on-rail movement mechanism, powered by an 1800 rpm DC motor coupled with a 60:1 reduction gearbox, provided a stable and controlled output speed of 30 rpm. Simultaneously, the rotational system — driven by a NEMA 34 stepper motor and pulley arrangement — delivered an optimized output speed of 6 rpm, enabling smooth, vibration-free angular positioning.

During the subsequent field-testing phase, the assembled platform continued to perform in alignment with its design expectations. The system not only retained its high accuracy across all axes but also demonstrated consistently smooth operation throughout extended test cycles. The depth control mechanism achieved precise vertical positioning, the rotational subsystem smoothly completed angular scans without noticeable lag or misalignment, and the linear on-rail movement maintained steady, synchronized travel. Real-time encoder feedback verified that all movements remained well within mechanical boundaries, and limit switches effectively triggered immediate cutoffs when tested at travel extremes. Field results confirmed that the actual movement profiles mirrored simulated performance curves, reinforcing the mechanical stiffness, alignment precision, and operational stability of the platform in real-world conditions.

With structural robustness, validated high positional accuracy, integrated safety mechanisms, and seamless remote control, the SONAR positioning platform successfully achieved all design and performance objectives. Its smooth operation and field-tested reliability signify its readiness for robust deployment in SONAR mapping, underwater scanning, and broader hydrographic survey applications.

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS

This is the final closing chapter of the project, highlighting the results and conclusion of the work done on the project along with some future recommendations.

6.1 Conclusion

This project successfully achieved the end-to-end design and manufacturing of a fully remote-controlled machine that meets industry standards while remaining feasible within the machinist-based manufacturing ecosystem of Pakistan. The system delivers high precision, achieving linear accuracy within 1 mm across a 10×6 m span and rotational accuracy of 1° across a full 360° range, made possible through the integration of stepper motors and high-resolution encoders.

Mechanical alignment, particularly the concentricity of assemblies such as the rack and flange was one of the most critical and successful aspects of the project, ensuring smooth rotation, minimizing eccentric motion, and reducing unnecessary loading on bearings. To enhance reliability, limit switches and mechanical stoppers were implemented as key safety features. These ensure the machine halts immediately in the event of communication failure or over-travel, protecting both the equipment and its surroundings.

Motion control was carefully managed using NEMA 34 stepper motors and motor drivers, with future considerations for DC motor integration for smoother rail movement. Structural integrity was maintained by fabricating precise linkages for Al extrusion alignment, effectively minimizing bending under operational loads. A hardware-based feedback loop using feedback motors was implemented to further refine accuracy and responsiveness. In addition, optimization of the winch bracket contributed to weight reduction and facilitated easier dismantling and re-alignment.

Overall, this project not only fulfills its intended functional and safety requirements but also showcases the potential for developing sophisticated, locally manufacturable automation systems with high precision and robust design.

6.2 Future Recommendations

Following are some recommendations that can improve the quality of the product.

6.2.1 Upgrade to Industrial-Grade Feedback Controllers

To further enhance accuracy and response time, future iterations should consider using advanced PID-based or industrial motion controllers.

6.2.2 Improve Communication Protocols

Implementing robust communication protocols (e.g., CAN or RS485) can further reduce the risk of data loss or lag, improving system responsiveness.

6.2.3 Introduce Modular Design for Portability

Designing the machine with modular sub-assemblies will improve portability, maintenance, and potential scalability for field deployment or industrial use.

6.2.4 Develop a Complete Troubleshooting Dashboard

Expand the LED-based troubleshooting system into a more comprehensive dashboard that can indicate specific failure points or suggest corrective actions.

6.2.5 Regular Maintenance Protocols

Implement a maintenance schedule and checklist, especially for aligned components and safety systems, to preserve long-term precision and safety.

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